# STAGE1:Generative COT Enhancement Engine

## step1 VQA Alignment

{"Q": "Has this leaf contracted Early Blight?",
"A": "No, this tomato leaf has Late Blight."}

**Processing method:**images resize to **384x384**; QA extract **18.93%** preserving original distribution;Shuffled into **single sequence**

## step2 Automatic CoT Generation.

**Prompt Template:**

Type detection (Diagnosis/Control)

Few-shot examples (3 demos)

**Step-by-step** guidelines

Hyperparams: T=0.3, max_tokens=1024

**Deepseek-VL2**

## step3 CoT Judge and Enhenced

**GPT-4**

**RESPONSE:<think>**Step 1: Analyze disease characteristics... Step 2: Recommend cultural/preventive practices ... **</think><answer>**No, this tomato leaf has Late Blight**</answer>**

**Guideline:**
Accuracy ;Completeness ;LogicalFlow ;Relevance ;
Clarity Threshold: **τ = 8.0/10.0**

# STAGE2: GRPO Reinforcement Learning

## step1 Smart Vocabulary System

**Plant Synonyms**
15 species x 5-8 variants

**Disease Synonyms**
20+ diseases x 3-6 variants

**Treatment Keywords**
4 categories, 47 terms

**5-tier fuzzy matching**
(1.0 → 0.25)

## Step2 GRPO Training Engine

**Dynamic Dual-Path Evaluation**

**Diagnosis**
99.6%
Plant+ Disease

**Treatment**
0.4%
4 Categories

**Qwen-2.5VL-3B**

**System Prompt:**
**Requirements**:Output as steps (Step 1, 2, 3, 4).Use only one step set per question type. Never mix.
**Step Guidelines（For Identification Questions）**：
Step 1: Identify host plant.Step 2: Describe symptoms. Step 3: Analyze disease pattern.Step 4: Provide diagnosis & confidence.
**Output Format**： **<think>**(Step reasoning) **</think><answer>**(Final Answer) **</answer>**

## Step3 Reward &Update

**Total Reward: R(o) = 0.5·R_f + 2.0·R_a + 0.5·R_r [0, 3.0]**

**Format R_f**
Structure:<think><answer>
Step count: 3-4 steps
Step quality (length)
Reasonable length

**Answer R_a (Main)**
Diagnosis (99.6%):
Plant match (0.8)
Disease match (1.2)
Control (0.4%):
Treatment keywords (2.0)

**Reasoning R_r**
Logical coherence (0.25)
Professionalism (0.15)
Completeness (0.10)

**Group Relative Advantage:**
num_generations = **3**(OOM prevention) ; advantage =(r-μ)/ σ ; Relative evaluation

**Clipped PPo Update:**
max grad norm = **0.3**(gradient explosion prevention) ; KL penalty + stability ;
DeepSpeed **ZeRO-3** (accelerate, balance); BF16 + grad_accum=4